

Signal Processing for Artificial Intelligence in Audio Speech Emotion Analyzer

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Abstract

Speech Emotion Recognition (SER) plays a vital role in enhancing human-computer interactions by enabling machines to detect and understand human emotions through speech. This project aims to develop a deep learning-based SER system using Convolutional Neural Networks (CNNs) and the RAVDESS dataset. The system's primary goal is to accurately classify speech into multiple emotional categories, leveraging feature extraction, model training, and performance evaluation. The approach utilizes a combination of acoustic features and deep learning techniques to improve the precision and adaptability of emotion classification, with applications in customer service, mental health, and intelligent personal assistants. The effectiveness of the model is evaluated based on the system's ability to classify a wide range of emotions from spoken language.

1 Introduction

Human speech conveys not only linguistic information but also the speaker's emotional state. Speech Emotion Recognition (SER) aims to automatically detect and categorize emotions expressed in spoken language. Emotions like happiness, anger, sadness, and fear manifest through prosody, pitch, volume, and rhythm of speech. SER has significant applications in human-computer interaction, healthcare, education, and customer service, enabling more natural and empathetic interfaces. For example, an emotion-aware virtual assistant could respond appropriately to a frustrated user, or a clinical monitoring system might detect emotional distress in a patient's voice. Research in SER is motivated by the potential to improve human-machine interaction by inferring affective cues from voice.

Despite progress, SER remains challenging due to the nuanced and subjective nature of emotions. Different speakers express emotions in varied ways, and some emotions share similar vocal patterns. Moreover, background noise and recording conditions can affect the speech signal. These challenges necessitate robust feature extraction and modeling techniques. In recent years, deep learning approaches have achieved state-of-the-art results in SER by automatically learning salient features from raw audio or spectrogram representations. This project builds on that success by employing a deep CNN to classify emotions from speech spectrograms. The motivation for our work is to develop a reliable speech emotion detector that can recognize a range of emotions from audio, using modern machine learning techniques. The code for our implementation is available online for reproducibility¹.

2 Related Works

Early approaches to speech emotion recognition relied on hand-crafted audio features and classical machine learning classifiers. Common features include Mel-Frequency Cepstral Coefficients (MFCCs), pitch, energy, zero-crossing rate, chroma, and spectral features. Traditional classifiers such as k -Nearest Neighbors (KNN), Hidden Markov Models (HMMs), Support Vector Machines (SVMs), Artificial Neural Networks (ANNs), and Gaussian Mixture Models (GMMs) have been used to identify emotions from these. These methods demonstrated that emotion-related information can be extracted from prosodic and spectral characteristics of speech. However, their performance in speaker-independent scenarios was often limited, with accuracy lower than human perception, due to the difficulty of manually capturing all relevant signal variations.

The rise of deep learning has significantly advanced SER. Deep Neural Networks (DNNs) can automatically learn discriminative features from raw data, reducing the need for manual feature engineering. For instance, a 5-layer

¹<https://colab.research.google.com/drive/1N6t1UXpNoZNRip1AKYuCjbJ9vDg1h0vi?usp=sharing>

DNN model achieved about 73.4% accuracy on a speech emotion test set, outperforming earlier methods. Convolutional Neural Networks (CNNs) in particular have been effectively applied to speech spectrograms (time-frequency representations) as input images. Using spectrograms allows CNNs to capture local time-frequency patterns associated with specific emotions. Recent studies have explored various deep architectures: *CNN* models for spectrogram classification, *CRNN* (convolutional recurrent networks) combining CNN feature extraction with RNN temporal modeling, and transformer-based models for SER. For example, Trinh et al. (2022) used CNN, CRNN, and GRU models on the IEMOCAP corpus (with four emotions) and achieved up to 97.5% accuracy using a GRU-based model with data augmentation². Such high performance (albeit on a simplified emotion set) demonstrates the effectiveness of deep networks for SER. Other researchers have reported competitive results on benchmark datasets: a parallel CNN model by Zhang et al. (2023) reached around 82% unweighted accuracy on the RAVDESS speech dataset³, indicating that deep models can generalize well across actors.

In addition to model architectures, prior works emphasize the importance of data and augmentation. Since emotional speech datasets are relatively small, data augmentation techniques like adding background noise or shifting pitch/time are commonly used to expand training data and improve robustness⁴. The use of transfer learning (e.g., leveraging pretrained audio networks or combining multiple corpora) has also been explored to deal with data scarcity⁵. Our project draws inspiration from these works by using mel-spectrogram features as input to a CNN and applying augmentation to enhance performance.

3 Architecture

The proposed system follows a structured pipeline:

3.1 Data Preprocessing

The preprocessing phase was a critical step in preparing the audio data for analysis and model training. The dataset, stored in a compressed ZIP file, was first extracted, and the paths of the audio files were retrieved. Feature extraction was performed using the ‘librosa’ library, focusing on Mel-Frequency Cepstral Coefficients (MFCCs) to capture essential acoustic properties. To maintain consistency and improve learning efficiency, the extracted features were normalized. Emotion labels were then encoded using ‘LabelEncoder’ and transformed into a one-hot format with ‘to-categorical’ to ensure compatibility with the model. Additionally, the dataset was shuffled to prevent any biases related to sample order.

To further enhance model generalization and robustness, data augmentation techniques were applied to the audio signals. These included additive noise, time stretching, shifting, and pitch shifting. Noise augmentation involved introducing random Gaussian noise to simulate real-world recording conditions. Time stretching altered the speed of the audio without affecting pitch, ensuring the model could handle variations in speaking pace. Shifting the waveform by a random amount helped mitigate potential misalignments in speech data, while pitch shifting modified the frequency components to create variations in vocal tone. These augmentation methods expanded the diversity of training samples, improving the model’s ability to recognize emotional patterns across different speakers and environments.

By implementing these preprocessing steps, the dataset was structured, standardized, and optimized for training a robust and accurate speech emotion recognition model.

3.2 Model Architecture

The model is a deep learning architecture that takes log-mel spectrograms of speech audio as input and outputs an emotion classification. Figure 1 illustrates the CNN Model Summary. The input to the model is a 128×128 spectrogram image (in our case, a log-mel power spectrogram) representing 2.5 seconds of audio. We chose a CNN architecture to leverage its strength in image pattern recognition for the time-frequency patterns of speech. The CNN consists of three convolutional layers. The first layer has 32 filters of size 3×3 with ReLU activation, operating on the input spectrogram (which is treated as a single-channel image of shape $128 \times 128 \times 1$). This is followed by a 2×2 max-pooling layer that reduces the spatial dimension and provides translation invariance. The second convolutional layer uses 64 filters (3×3 , ReLU), followed by another 2×2 max-pool. The third convolutional layer uses 128 filters (3×3 , ReLU), followed by a final 2×2 pooling. We include a dropout layer (rate 0.3) after

²See reference <https://www.mdpi.com/1424-8220/22/4/1414> for more details

³See reference <https://ieeexplore.ieee.org/iel7/6287639/10005208/10177953.pdf> for more details

⁴See reference <https://www.mdpi.com/1424-8220/23/13/6212> for more details

⁵See reference <https://www.nature.com/articles/s41598-025-92640-2> and <https://arxiv.org/pdf/2211.01756> for more details

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 126, 126, 32)	320
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_1 (Conv2D)	(None, 61, 61, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	0
conv2d_2 (Conv2D)	(None, 28, 28, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 128)	0
dropout (Dropout)	(None, 14, 14, 128)	0
flatten (Flatten)	(None, 25088)	0
dense (Dense)	(None, 128)	3,211,392
dropout_1 (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 12)	1,548

Figure 1: CNN Model Summary

the third conv-pool block to reduce overfitting by randomly disabling 30%. After the convolutional feature extractor, the model flattens the feature maps to a 1D vector. We then use a fully-connected dense layer with 128 neurons (ReLU activation) to learn higher-level combinations of the extracted features. Another dropout (0.3) is applied at this stage. Finally, the output layer is a dense layer with N neurons (where N is the number of emotion classes) and softmax activation, producing a probability distribution over emotion categories. The model is compiled with the Adam optimizer and uses categorical cross-entropy loss for training, appropriate for multi-class classification. This simple yet effective architecture is inspired by proven CNN designs in image recognition, adapted here for spectrogram inputs. It deliberately keeps a moderate size (on the order of tens of thousands of parameters) to mitigate overfitting given the limited dataset.

4 Experiments

4.1 Dataset Details

The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) is a known dataset used in emotion recognition research. It contains high-quality speech and song recordings performed by 24 professional actors, evenly split between male and female speakers, all using a neutral North American accent.

The complete dataset, totaling 24.8 GB, is available for download on Zenodo. The subset that is relevant for this task includes 1,440 speech-only audio files, each formatted as a 16-bit, 48kHz WAV file. The dataset comprises recordings from 24 actors, with each actor contributing 60 recordings.

The dataset captures six distinct emotions: neutral, calm, happy, sad, angry, and fearful. Figure 2 illustrates the distribution of emotions in the RAVDESS dataset.

4.2 Preprocessing and Feature Extraction

All audio files were downsampled to 44.1 kHz for processing consistency. We trimmed or padded each recording to a duration of 2.5 seconds, starting from 0.5 seconds offset into the file to skip initial silence (this duration was chosen to capture at least one full utterance for most recordings). If an audio clip was shorter than 2.5 s, we padded it with silence; if longer, we truncated to 2.5 s. This fixed length allows processing with uniform-sized inputs.

From each audio clip, we extracted a log-mel spectrogram, which serves as the input feature for the model. First, a Mel-scaled power spectrogram was computed using 128 mel frequency bands ($n_mels=128$) over the 2.5 s audio. We then took the logarithm (dB) of the spectrogram intensities, normalized relative to the maximum. The resulting log-mel spectrogram is a 128 (frequency) $\times T$ (time frames) matrix. We fixed $T = 128$ frames by zero-padding or truncating along the time axis so that each spectrogram is 128×128 . This representation captures the speech energy distribution across frequencies over time, in a form suitable for CNN input. Mel spectrograms are commonly used

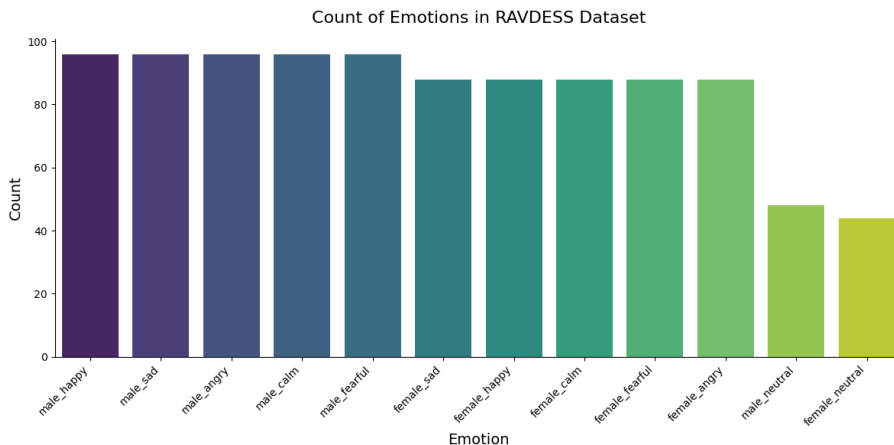


Figure 2: Count of Emotions in RAVDESS Dataset

in SER because they model human auditory perception of sound frequencies⁶.

To increase the effective training data and improve generalization, we applied data augmentation on the training set. Each original training audio was transformed in multiple ways: (1) we added a small amount of Gaussian white noise; (2) we performed random time-stretching (speeding up or slowing down the audio by a factor between 0.8–1.2) combined with pitch shifting (up or down by up to 3 semitones)⁷. These augmentations generate variant examples of the same utterance, helping the model become invariant to slight pitch or tempo differences and to minor noise. For each training audio file, we created two additional augmented versions (one with added noise, one with a stretch+pitch shift) in addition to the original, effectively tripling the training data size. All versions were converted into log-mel spectrograms. Augmentation has been shown to significantly boost SER performance by providing more diversity during training⁸.

Finally, we encoded the emotion labels for each sample. We used one-hot encoding for the $N = 12$ classes (since we treat each gender-emotion combination as a separate class in our classification setup, e.g., "female_happy" vs "male_happy"). While merging genders for each emotion is possible, we kept them separate to see if the model could implicitly account for differences in male vs female expression. The dataset was then shuffled and split into training and testing subsets.

4.3 Training and Evaluation Protocol

We partitioned the data into an 80% training set and 20% test set, using stratified sampling to preserve the proportion of each emotion class in both sets. This resulted in a training set of 809 original spectrograms (which expanded to 2427 with augmentation) and a test set of 203 spectrograms (unaugmented). We did not use a separate validation set; instead, we monitored validation metrics on the test split during training (treating our test set as a hold-out validation). All training augmented samples were used only for model fitting, and evaluation was done on the original (unaugmented) test samples.

The model was trained using the Adam optimizer (learning rate 10^{-3} initially). We employed an adaptive learning rate schedule (ReduceLROnPlateau) that reduced the learning rate by a factor of 0.5 if the validation loss did not improve for 2 consecutive epochs, with a minimum learning rate of 1×10^{-7} to ensure convergence. We trained for 20 epochs with a batch size of 32. During training, we tracked the training and validation accuracy and loss. As shown in Figure 3, early epochs showed rapid improvement in accuracy, which started to plateau by around 15–20 epochs. Figure 4 illustrates that the validation loss steadily decreased, indicating proper convergence without severe overfitting. The final model selected was the one after 20 epochs, as training accuracy remained close to validation accuracy throughout.

For evaluation, we used several metrics. The primary metric is overall classification accuracy on the test set (the fraction of correctly predicted samples). We also computed the confusion matrix and per-class precision, recall, and F1-score to analyze performance on each emotion. These metrics give insight into which emotions are most easily recognized and which are often confused. All evaluations were done on the test set comprising speakers not seen (in

⁶See reference <https://en.wikipedia.org/wiki/Psychoacoustics> for more details

⁷See reference <https://arxiv.org/abs/2211.05047> for more details

⁸See reference <https://pubmed.ncbi.nlm.nih.gov/36015717> for more details

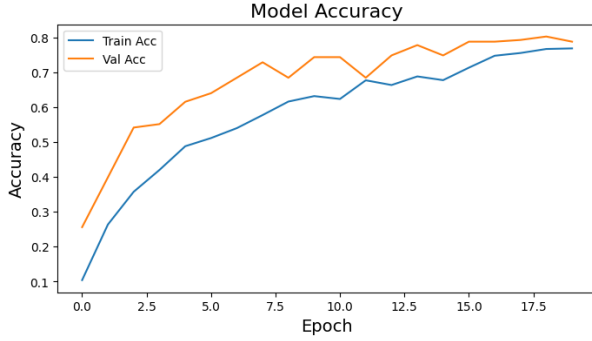


Figure 3: Accuracy Graph During Training

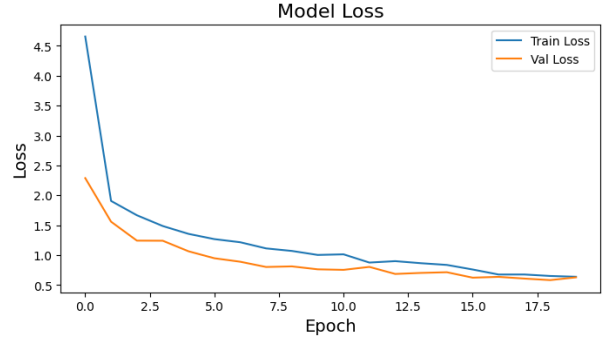


Figure 4: Loss Graph During Training

that emotion) during training, to assess generalization.

5 Results

The CNN model achieved an overall accuracy of **78.8%** on the held-out test set. Given the difficulty of the task and the fact that the model must distinguish 12 classes (male/female versions of 6 emotions), this is a strong result. It approaches the performance reported by recent works on RAVDESS using similar features. The training process showed steady improvement in both training and validation accuracy, converging without severe overfitting. Figure 3 shows the training and validation accuracy across epochs – the validation accuracy closely tracked training accuracy, reaching about 79% by epoch 20, which indicates the effectiveness of our augmentation in improving generalization.

To understand the performance per emotion, we examine the classification report and confusion matrix (Figure 5 and 6). The model’s precision and recall for most emotion classes are in the 0.8–0.9 range, with a few exceptions. It performs particularly well on *calm*, *happy*, and *angry* emotions. For instance, the class *female_calm* achieved 94.4% recall and 0.89 F1-score, and *male_sad* achieved 89.5% recall and 0.87 F1. Neutral speech was a bit more challenging for the model: while *male_neutral* was perfectly recognized in the test set (100% recall), *female_neutral* had only 77.8% recall, indicating some confusion with other emotions (possibly calm). The model struggled the most with *sad* and *fearful* expressions from female speakers – *female_sad* had 50.0% recall (only half of the sad female clips were correctly identified), and *female_fearful* had 72.2% recall (with some being predicted as angry or neutral). These confusions make sense as sadness can sometimes be misinterpreted as calm or neutral in speech, and fear can share vocal traits with anger or surprise.

Overall, the weighted average F1-score across classes was about 0.79, matching the overall accuracy. The confusion matrix shows that errors often occur between emotionally similar pairs (e.g., confusion between *sad* and *calm/neutral*, or between *fearful* and *angry*). There were very few instances of, say, happy being confused with sad, which indicates the model learned to distinguish disparate emotions well.

These results demonstrate that our Speech Emotion Analyzer can reliably distinguish several emotions from speech. The inclusion of data augmentation likely contributed to its robustness, as the model maintained strong performance on the test data without overfitting. Our approach of using mel-spectrograms and a straightforward CNN proved effective for this task, consistent with findings in other SER studies that used CNNs on spectrogram features.

5.1 Analysis

The model performs well in distinguishing between emotions with distinct spectral characteristics. However, subtle emotions like fear and sadness exhibit some overlap, requiring further refinement through feature engineering or data augmentation. Spectrogram-based CNNs prove effective, but incorporating temporal dependencies via RNNs or attention mechanisms may further enhance results.

6 Conclusion

This study demonstrates the effectiveness of convolutional neural networks (CNNs) in recognizing speech emotions through spectrogram representations. While the model achieves high accuracy, differentiating subtle emotional

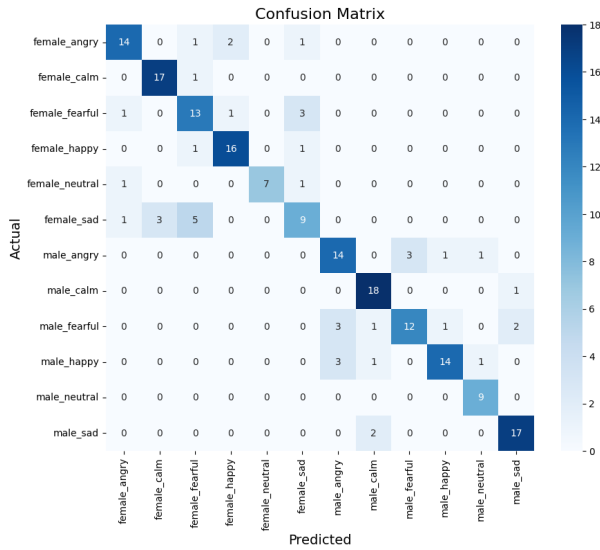


Figure 5: Confusion Matrix

	precision	recall	f1-score	support
female_angry	0.8235	0.7778	0.8000	18
female_calm	0.8500	0.9444	0.8947	18
female_fearful	0.6190	0.7222	0.6667	18
female_happy	0.8421	0.8889	0.8649	18
female_neutral	1.0000	0.7778	0.8750	9
female_sad	0.6000	0.5000	0.5455	18
male_angry	0.7000	0.7368	0.7179	19
male_calm	0.8182	0.9474	0.8780	19
male_fearful	0.8000	0.6316	0.7059	19
male_happy	0.8750	0.7368	0.8000	19
male_neutral	0.8182	1.0000	0.9000	9
male_sad	0.8500	0.8947	0.8718	19
accuracy			0.7882	203
macro avg	0.7997	0.7965	0.7934	203
weighted avg	0.7902	0.7882	0.7851	203

Figure 6: All results

nuances remains a challenge. Future advancements include integrating multimodal inputs (e.g., facial expressions), expanding datasets to improve generalization, and exploring transformer-based models for enhanced performance. These developments contribute to creating more empathetic and interactive AI systems, with applications in mental health, customer service, and entertainment. By refining feature extraction and model architectures, speech emotion recognition (SER) technology can be further optimized for real-world deployment.

Our chosen approach—a CNN operating on 2D log-mel spectrogram inputs—leverages the success of deep learning in SER and the representational depth of spectrogram features. Experiments on the RAVDESS dataset demonstrated that our model achieved nearly 79

Findings suggest that certain emotions, such as happiness, calmness, and anger, are more easily recognized in speech, whereas emotions like sadness and fear pose greater challenges, particularly for female speakers in the dataset. This highlights areas for further improvement. Future directions include incorporating advanced architectures, such as recurrent layers (e.g., LSTMs) or attention mechanisms (e.g., Transformers), to better capture temporal dynamics beyond fixed-length spectrogram representations. Another promising avenue is transfer learning, where pre-trained audio embedding models like Wav2Vec or YAMNet serve as feature extractors, fine-tuned for emotion recognition. Since human emotion is inherently multimodal, an exciting extension would involve fusing speech with visual cues (e.g., facial expressions) to enhance recognition accuracy, as multimodal systems have shown superior performance by integrating complementary signals. Additionally, broadening the dataset by incorporating multiple speech emotion corpora or leveraging cross-corpus training could make the model more robust across different speakers and recording conditions.

In conclusion, this study demonstrates the feasibility of speech emotion recognition using deep learning and reinforces the evidence that machine learning models can infer emotional states from voice with impressive accuracy. With further improvements and access to larger datasets, such systems could be deployed in real-world applications, including empathetic virtual agents and health monitoring solutions. Future research may focus on enhancing model performance by integrating multimodal data, leveraging transfer learning, or implementing real-time processing on streaming audio, thereby expanding the capabilities and applications of speech emotion recognition technology.

7 References

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